

Question No:5

Problem Description

You are given a string which comprises of lower case alphabets (a-z), upper case alphabets (A-Z), numbers, (0-9) and special characters like !,.,; etc.

You are supposed to find out **which character occurs the maximum number of times and the number of its occurrence**, in the given string. If two characters occur equal number of times, you have to output the character with the lower **ASCII value**.

For example, if your string was: **aaaaAAAA**, your output would be: **A 4**, because **A** has lower ASCII value than **a**.

Input format:

The input will contain a string.

Output format:

You've to output two things which will be separated by a space:

- i) The character which occurs the maximum number of times.
- ii) The number of its occurrence.

Constraints:

The maximum length of the string can be 1000.

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

puppy is a dog!!!!!!!!!!!!

EXPECTED OUTPUT

! 12

Test Case 2

INPUT (STDIN)

leana is a singer!!!!!!!

EXPECTED OUTPUT

! 7

Code:

```
1  #include <stdio.h>
2  #include <string.h>
3  int main() {
4      char str[101];
5      int freq[256] = {0};
6      fgets(str, sizeof(str), stdin);
7
8      int n = strlen(str);
9
10     if (n > 0 && str[n - 1] == '\n') {
11         str[n - 1] = '\0';
12     }
13
14     for (int i = 0; str[i] != '\0'; i++) {
15         freq[(unsigned char)str[i]]++;
16     }
17
18     int max = 0;
19     char ch;
20
21     for (int i = 0; i < 256; i++) {
22         if (freq[i] > max) {
23             max = freq[i];
24             ch = (char)i;
25         }
26     }
27     printf("%c %d\n", ch, max);
28     return 0;
29 }
```

Output:

Test Case-1:

```
puppy is a dog!!!!!!!!!!!!!!
! 12
PS C:\Users\kesha\OneDrive\Documents\3sem\
```

Test Case-2:

```
leana is a singer!!!!!!!
! 7
PS C:\Users\kesha\OneDrive\Documents\3sem\DSA\DSA Practi
```

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